Data Analytics Internship – Task 2

Sales Performance Analysis

Submitted to: Alfido Tech

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Domain: Data Analytics Internship
Tools Used: Python, Pandas, Matplotlib, Google Colab

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1. Introduction

This report presents a detailed analysis of sales performance using the Superstore dataset. The objective of this task is to evaluate sales across different regions, product categories, and time periods. The analysis also identifies top and low-performing products and highlights seasonal trends in sales.

The insights generated from this analysis help in understanding business performance and provide actionable recommendations to improve sales and profitability for Alfido Tech.

2. Dataset Description

The dataset contains order-level information including:

- Order Date
- Ship Date
- Region
- Category
- Product Name
- Sales

This dataset is used to analyze revenue generation, customer demand, and product performance.

3. Data Cleaning

The following steps were performed:

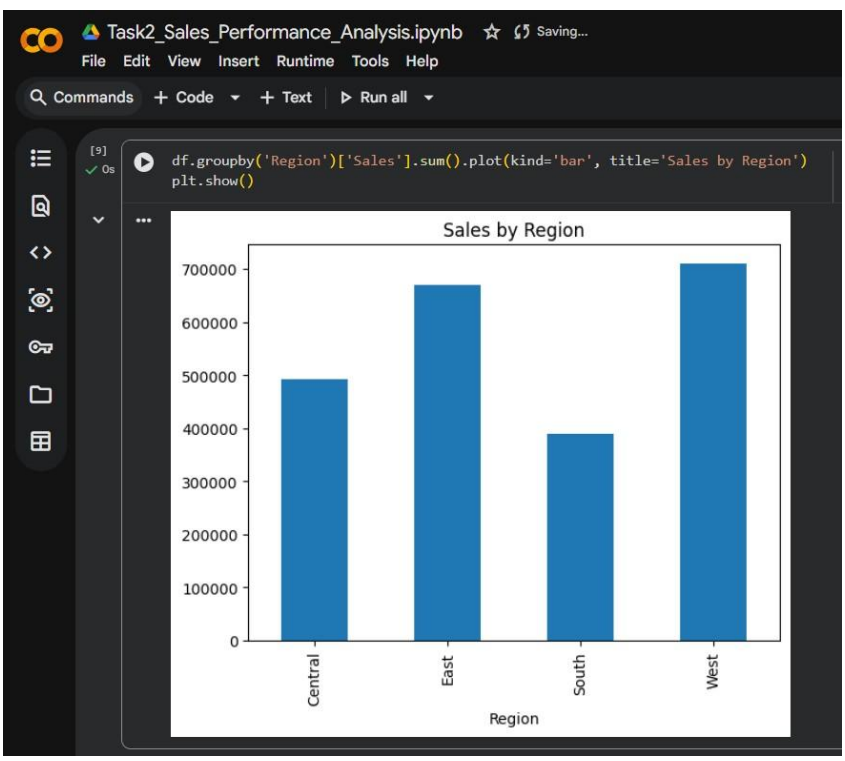
- Removed duplicate records
- Converted Order Date and Ship Date into datetime format
- Extracted Year and Month for trend analysis
- Checked and handled missing values

4. Key Performance Indicators (KPIs)

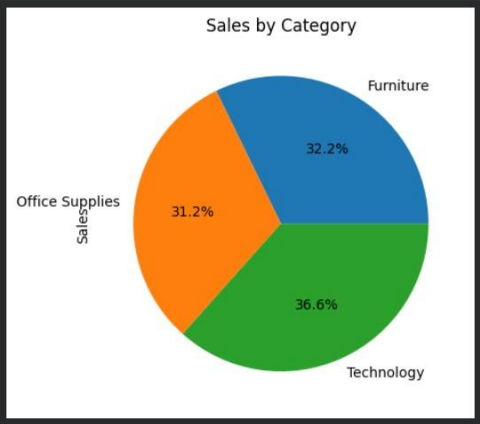
```
[8]
0s
print("Total Revenue:", df['Sales'].sum())
print("Average Order Value:", df['Sales'].mean())
if 'Profit' in df.columns:
    print("Total Profit:", df['Profit'].sum())
if 'Profit' in df.columns:
    total_profit = df['Profit'].sum()
    total_sales = df['Sales'].sum()
    print("Profit Margin (%):", (total_profit / total_sales) * 100)

Total Revenue: 2261536.7827000003
Average Order Value: 230.7690594591837
```

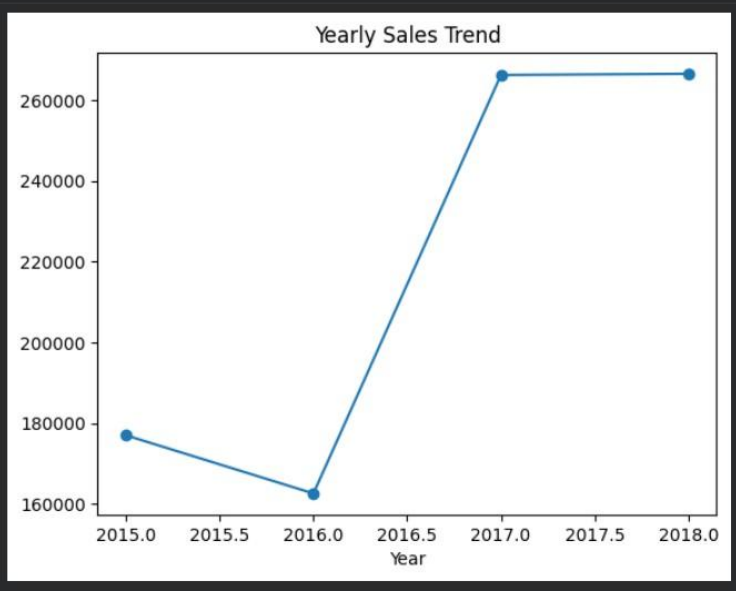
5. Data Visualization



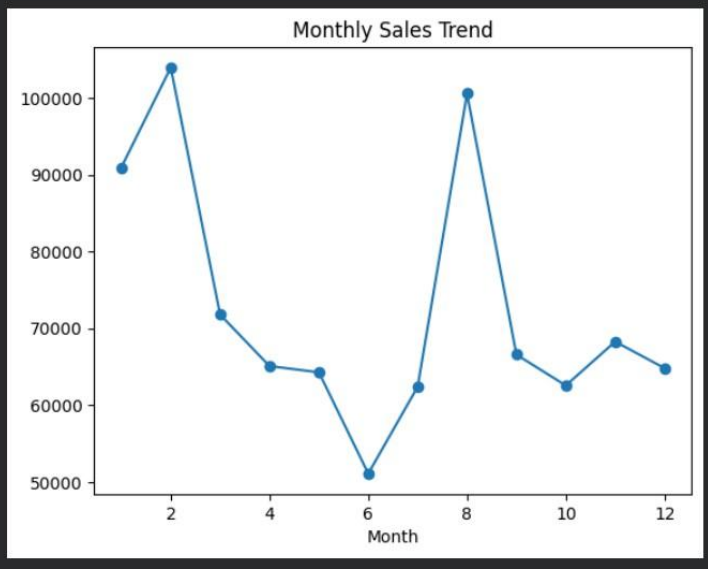
```
[10]
df.groupby('Category')['Sales'].sum().plot(kind='pie', autopct='%1.1f%%', title='Sales by Category')
plt.show()
```



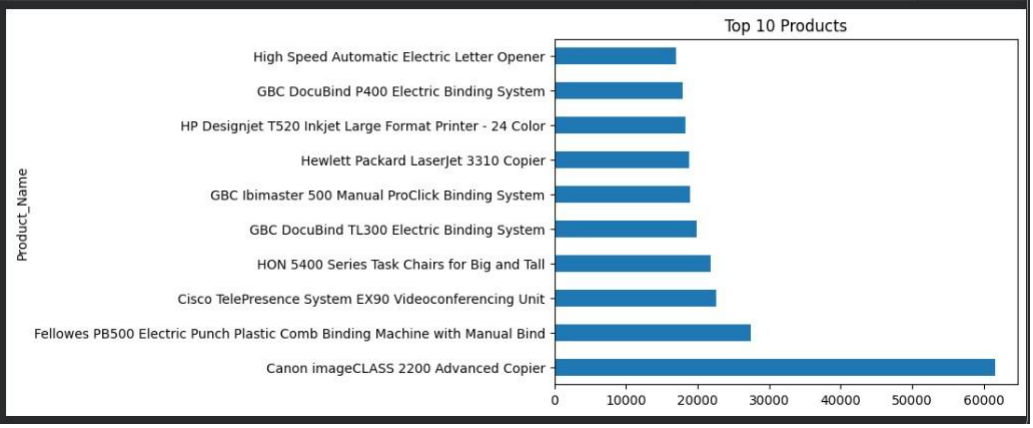

```
[11]
df.groupby('Year')['Sales'].sum().plot(marker='o', title='Yearly Sales Trend')
plt.show()
```



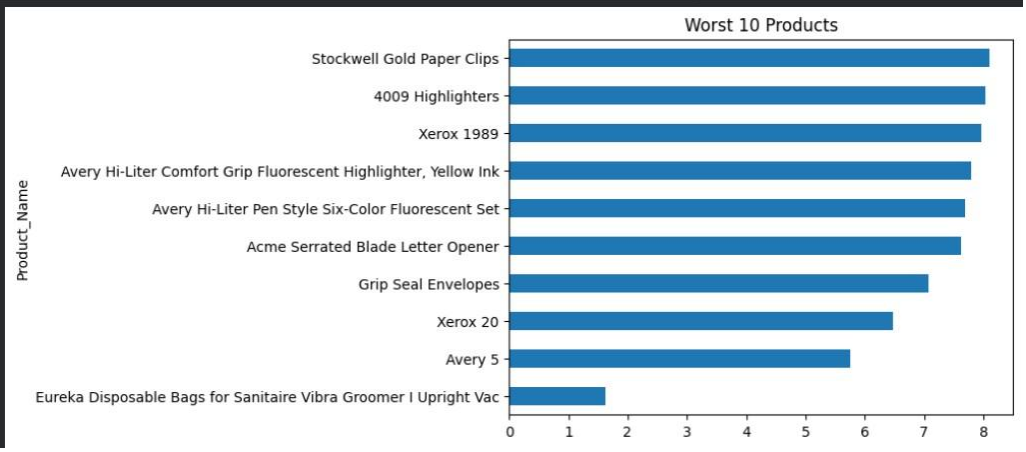
```
[12]
df.groupby('Month')['Sales'].sum().plot(marker='o', title='Monthly Sales Trend')
plt.show()
```




```
df.groupby('Product_Name')['Sales'].sum().sort_values(ascending=False).head(10).plot(kind='barh', title='Top 10 Produ
plt.show()
```




```
df.groupby('Product_Name')['Sales'].sum().sort_values().head(10).plot(kind='barh', title='Worst 10 Products')
plt.show()
```



6. Key Insights

Certain regions contribute significantly higher revenue than others

Technology category shows strong overall performance

Sales demonstrate a consistent upward trend over time

Seasonal peaks are observed in specific months

A small number of products contribute to a large portion of total sales

7. Recommendations

1. Focus marketing efforts on high-performing regions
2. Increase inventory and promotion of top-selling products
3. Improve or remove low-performing products
4. Launch targeted campaigns during peak sales months
5. Optimize pricing strategies to improve profit margins

8. Executive Summary

This project analyzed sales data to identify patterns and trends. Visualizations helped understand category performance and regional distribution. Recommendations were provided to improve business strategies.

9. Project Links

https://github.com/Swayamchavan/InternSpark-Internship